

Birla Institute of Technology & Science, Pilani, Rajasthan
First Semester 2020-2021
Comprehensive Exam-Part A (14-12-2020)

Course: EEE F311 Communication Systems

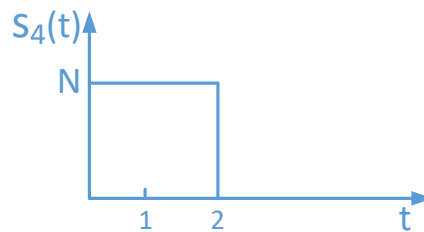
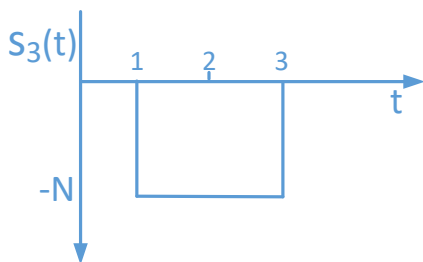
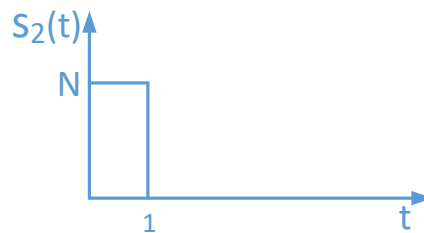
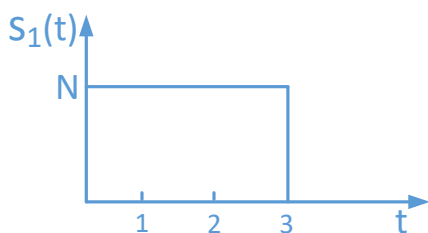
Duration: 55 Minutes, Marks: 30

There are 3 question each of 10 marks.

Instructions: Write down your solutions by substituting numerical value of N (explicitly write N in a box) at the start of writing the solution of each question. Any solution in terms of N will not be evaluated. Write sufficient steps to arrive at the solution.

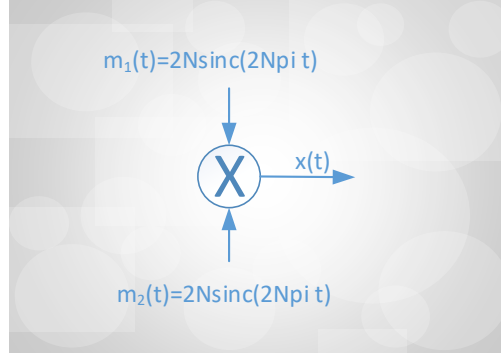
1. N = Last digit of your BITS ID. If the last digit is zero, use $N = 5$.

- (a) An FM signal given as $x(t) = A \cos(2\pi f_c t + Nt^3)$ is passed through a PLL to track its instantaneous phase with a second-order loop filter $H(s) = 1 + \frac{1}{s} + \frac{N}{s^2}$. Find the steady state error of the instantaneous phase. Use standard notations for other parameters of the PLL.
- (b) Find a set of basis signals of $s_1(t)$, $s_2(t)$, $s_3(t)$, and $s_4(t)$ as shown in figure below and represent the signals $s_1(t)$, $s_2(t)$, $s_3(t)$, and $s_4(t)$ in terms of the basis signals.



2. N = Sum of last two digits of your BITS ID. If the sum of last two digits is zero, use $N = 12$.

- (a) A binary line-coded baseband PAM signal has the form $y(t) = \sum a_k p(t - kT_b)$ where a_k is $+N$ for bit '1' and 0 for bit '0' with equal probability, and the pulse $p(t) = 2N \text{sinc}(2N\pi t)$. Find the PSD of $y(t)$ for a random bit stream.
- (b) Find the signaling rate T_s of the signal $x(t) = m_1(t)m_2(t)$ (as shown in figure below) that satisfies the Nyquist criteria in the frequency domain $\sum_{m=-\infty}^{\infty} X(f - \frac{m}{T_s}) = T_s$.



3. $N =$ maximum of last two digits of your BITS ID. If maximum comes to be zero or 1, take $N = 6$.

Let an amplitude modulated signal be $x(t) = P \cos[600\pi t] + Q \cos[560\pi t] + Q \cos[640\pi t]$. Let the carrier power is $10N$ Watt and power efficiency is $5N\%$. Find the following (Your answer should not be in terms of P and Q).

- Express the modulated signal $x(t)$ in the form of $(A + m(t)) \cos 2\pi f_c t$ and then find the modulation index.
- Find the bandwidth of the FM modulated signal if $A + m(t)$ of part (a) is passed through an FM modulator with $k_f = 2N$ Hz/Volt and a FM carrier signal $N \cos 10000\pi t$.
- Find the modulation index if the FM signal of part (b) is passed through a square device.
- Find the bandwidth of the PM modulated signal if $A + m(t)$ of part (a) is passed through a PM modulator with $k_p = 2N$ rad/Volt and a PM carrier signal $N \cos 10000\pi t$.
- Find the SNR if an additive noise of power N is added to the PM modulated signal of part (d).

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Comprehensive Exam-Part B (14-12-2020)

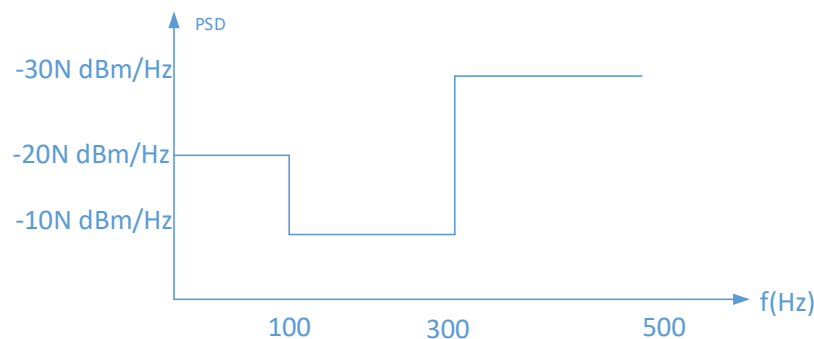
Course: EEE F311 Communication Systems

Duration: 75 Minutes, Marks: 40

There are 4 question each of 10 marks with each sub part of 5 marks.

Instructions: Write down your solutions by substituting numerical value of N (explicitly write N in a box) at the start of writing the solution of each question. Any solution in terms of N will not be evaluated. Write sufficient steps to arrive at the solution.

1. (a) Generate (7,4) linear systematic block code words (all 16 code words) where the generator matrix is given as $\mathbf{G} = [\mathbf{I}_{4 \times 4} \ \mathbf{P}_{4 \times 3}]$. Using the generated code words, find the error correcting capability of the code. The first column of the matrix $\mathbf{P}$ is the binary equivalent of the third last digit of your BITS ID. The second column of the matrix $\mathbf{P}$ is the binary equivalent of the second last digit of your BITS ID. The third column of the matrix $\mathbf{P}$ is the binary equivalent of the last digit of your BITS ID.
(b) Consider a $(N+2, N)$ linear systematic block code where the parity check digits is given as $c_{k+1} = d_1 + d_2 + d_3 + \dots + d_k$ and $c_{k+2} = d_1 + d_2 + d_3 + \dots + d_k$. Find the code word for a data word consisting of N 1's.
 N = Last digit of your BITS ID. If the last digit is zero, use $N = 6$.
2. (a) N = Maximum of even digits of the last three digit. If $N = 0$ or $N = 2$, take $N = 4$.
 N_1 = Maximum of even digits of the last three digit. If $N_1 = 1$, take $N = 5$.
A discrete channel has two inputs x_1 and x_2 and two outputs y_1 and y_2 . The probability of x_1 is $P(x_1) = \frac{1}{N}$ and transition probabilities are $P(y_1|x_1) = \frac{1}{N_1}$ and $P(y_1|x_2) = \frac{1}{N_1+2}$. Write the transition matrix, find the conditional entropy $H(Y|X)$, and find the mutual information of the channel.
(b) The power spectral density of a transmit signal for a channel bandwidth 500 Hz is shown in the figure. Find the channel capacity if double-sided noise spectral density is $-20N_1$ dBm/Hz.



3. N = Sum of all digits of your BITS ID including year and branch code.

A general digital transmission model with additive noise is $y = x + n$ where $x = \pm N$ with unequal probability $P(x = -N) = P_1$ and $P(x = +N) = P_2$. To analyze the probability of error, a general rule for threshold detection is $\frac{P_1 f(x|N)}{P_2 f(x|-N)} \geq 1$, where $f(x)$ denotes the probability distribution function.

- If the additive noise n has a distribution $f(n) = \frac{1}{\sqrt{2}} e^{-\sqrt{2}|n|}$, find the average SNR and average BER. Your final answer should be numeric.
- If the additive noise n is Gaussian distribution with mean 0 and variance σ^2 , find the average SNR and average BER (in terms of Q-functions).

4. N = Sum of last three digits of your BITS ID.

Let signal $s(t)$ is used to transmit bit '0' and signal $-s(t)$ is used to transmit bit '1' over a channel $H(f)$ as shown in figure below. The double sided AWGN noise PSD is $N_0/2 = 2N$.

- If the switch for channel is connected to (1), $H(f) = 1$, then find the average SNR at the output of the match filter signaled at T_b (i.e., using the signal $r(T_b)$).
- If the switch for channel is connected to (2), $H(f) = (1 + \cos 2\pi f T_b) e^{-j2\pi f N}$, then find the MF $f(t)$ and the average BER in terms of Q-function.

